Problem Statement-

Every year, ERC's freshman orientation has one special aspect- the hands-on activity. Your job now is to design a new activity for next year's orientation. LED flasher, a clap switch, or a tone generator - What are you gonna come up with? Make sure it stays budget friendly as we give out about 300 kits for this activity, and keep it simple- not everyone is an electronics geek. Prices of components can be found on Robu.

Submit a Bill of Materials (BOM) with approximate costs (refer Robu or similar sources), a circuit diagram or connection layout, an explanation of its working and so that anyone can build and understand the project independently.

Proposal-

Touch Sensor Switch-

A mini electronics project, that toggles an LED on and off, when a person touches a bare wire. It has no switch neither a button- just a finger. The main reason behind opting for this activity is because it has a wow-factor, can be built in less than 30 minutes and is very beginner friendly.

Technically, it is a capacitative touch sensitive on/off switch without mechanical contact. The body and the wire form a natural capacitative coupler in this case.

Major Components required for this activity are as follows:

1. NE555
2. Resistors(1kohm and 10kohm)
3. LED
4. Touch wire(a bare jumper wire acting as a capacitative sensor)

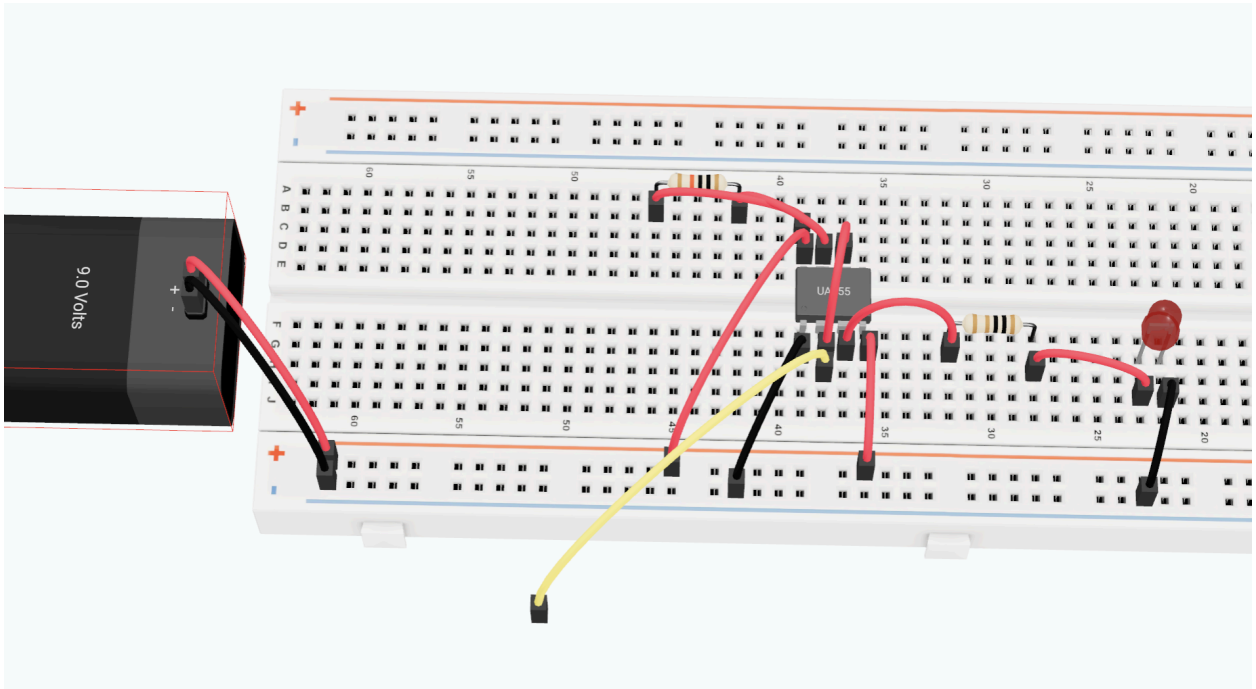
Circuit:

I modelled the circuit(next page) online using a website called withDiode, which helps you to simulate circuits. The yellow wire in this case is the 'touch wire', the black wires are for connections to the ground. There are two resistors, 1kohm connecting pin3 to the LED and 10kohm connecting pin7 and pin 8.

Working:

- Initially the LED is off, when you touch the wire, since your body has stray capacitance to Earth ground, your body gets electrically connected to pin2.
- The 50 Hz AC electric field permeating the room (from wall wiring, power supplies etc.) constantly induces a small AC voltage on your body, this gets coupled directly to pin2 via your body capacitance.

- On the negative half-cycle of that 50 Hz noise, Pin 2 momentarily drops below ~3V, which after a series of events inside the NE555 increases the voltage output to pin3 (upto ~8V), thus switching on the LED.
- And when you touch the wire again, the same cycle happens but this time in the positive half cycle of the 50Hz noise, which increases the output to pin2 momentarily (upto ~ 6V).
- Thus pin3 output goes low ~0V, switching off the LED.



Bill of Materials:

Component	specs	quantity	Unit price	Kit cost
NE555 timer IC	DIP- 8	1	5 ruppee	5 ruppee
LED(red)	20mA	1	2 ruppee	2 ruppee
1kohm resistor	0.25W	1	0.5 ruppee	0.5 ruppee
10kohm resistor	0.25W	1	0.5 ruppee	0.5 ruppee
Half Breadboard	400 tie- points	1	35 ruppee	35 ruppee
Jumper wires	Male to male	10	1 ruppee	10ruppee

9V battery snap	With bare wires lead	1	10 ruppee	10 ruppee
			TOTAL	63 ruppee/kit
			For 300 kits	18,900 ruppee

So, the approximate cost of conducting this activity is ~18,900 ruppees.